

Is the pre-hallucination signal *different* from the in-hallucination signal?

Evidence from nine probe-based tests on $RAGTruth \times \{Mistral-7B-Instruct-v0.1, Llama-2-13b-chat-hf\}$.

Test	SAME predicts	DIFFERENT predicts	Mistral (L=3)	Llama (L=1)	Verdict
Direction: does the probe vector point the same way at h_t and h_{t-1}?					
E1. $\cos(w_{k=0}, w_{k=1})$	$\approx$ bootstrap floor	≈ 0	0.652	0.514	SAME
E7b. Bootstrap cosine noise floor at h_t alone	calibrates E1 (cross-pos $\approx$ floor)	floor > > cross-pos	$0.631 \approx 0.652$	$0.525 \approx 0.514$	SAME
E9. Smallest principal angle (10-boot subspaces)	cross-pos $\approx$ same-pos floor	cross-pos > > same-pos floor	29.6° vs 25.9/26.8	35.8° vs 32.1/32.7	SAME
Content: does h_{t-1} carry information not in h_t (or vice versa)?					
E2. Concat $[h_t h_{t-1}]$ AUC gain vs best single	≈ 0	$\geq +0.03$	-0.004	-0.007	SAME
E3. Pearson($\sigma(w_A \cdot h_t), \sigma(w_B \cdot h_{t-1})$)	high (correlated)	low (independent)	0.735	0.573	SAME
E4. Difference probe AUC on $(h_t - h_{t-1})$	≈ 0.50	≥ 0.60	0.537	0.538	SAME
E5. Residual probe AUC ($h_{t-1} - \text{proj}_{w_A}$)	drops to ≈ 0.50	stays $\approx$ AUC(B)	0.837 (B=0.833)	0.750 (B=0.751)	MULTI
Subspace: geometric structure of the encoding					
E6. Cross-position probe transfer matrix	nearly uniform	diagonal-dominant	uniform; $k_{\text{train}}=0$ row best at every k_{eval}	uniform; same pattern	SAME
E8. Effective rank (AUC after 10 peels)	rank-1: collapse to 0.50	multi-D: slow decay	k=0: 0.789 (-0.05) k=1: 0.776 (-0.06)	k=0: 0.749 (-0.008) k=1: 0.740 (-0.011)	MULTI

Aggregate verdict
SAME signal, redundantly encoded in a high-dimensional residual-stream subspace, approximately position-invariant up to LR's data-resampling noise. The smooth lag-decay in the main study is signal attenuation within the shared subspace, not rotation into a different one.